Answer Key - Computer Science - Undergraduate Level Assessment 25 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
   Options: A) a; B) Chemistry; C) 0; D) 1
   Note: Correct option: B - Chemistry

2. **Answer:** A
   Options: A) O(1); B) O(n); C) O(n^2); D) O(log n)
   Note: Correct option: A - O(1)

3. **Answer:** A
   Options: A) n^m; B) n!/(n - m)!; C) n!; D) n!/(m!(n - m)!)
   Note: Correct option: A - n^m

4. **Answer:** A
   Options: A) ['Hi!', 'Hi!', 'Hi!', 'Hi!']; B) ['Hi!'] * 4; C) Error; D) None of the above.
   Note: Correct option: A - ['Hi!', 'Hi!', 'Hi!', 'Hi!']

5. **Answer:** D
   Options: A) 53; B) 29; C) 50; D) 100
   Note: Correct option: D - 100

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

7. **Answer:** False
   Options: A) True; B) False
   Note: LLM-generated false statement. Correct answer: 'All bit strings with more ones than zeros'.

8. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
   Options: A) True; B) False
   Note: LLM-generated true statement based on MCQ.

10. **Answer:** True
    Options: A) True; B) False
    Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** def adjac(ele, sub = []): | def get_coordinates(test_tup): | return (res)
    Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** def is_Monotonic(A): | return (all(A[i] <= A[i + 1] for i in range(len(A) - 1)) or
    Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key